KNOWLEDGE GRAPH BUILDER · GP REASONING ENGINE · 01 / 06

Turning any data into decision intelligence

One framework. Any client. Any data source. A living knowledge graph — ready for AI reasoning.

A knowledge graph connects everything your business knows

Products, markets, suppliers, competitors, warranties — all linked, queryable, and ready for AI agents to reason over. Instead of digging through reports, you ask questions and get answers backed by structured evidence.

Any data

Snowflake, Databricks, Oracle, SQL databases, PDF documents, CSV files, plain text, web data, GP workflow outputs — all ingested into one unified graph.

Client-agnostic

One framework — Rheem today, CoAction, MSL, MSIG and beyond. New clients onboarded in days, not months, with a single configuration file.

Ontology-driven

Our IP encodes the full semantic model: *class hierarchies* (how concepts nest), *object properties* (how entities relate), and *data properties* (what attributes each carries).

HOW IT WORKS · 02 / 06

From raw data to a connected knowledge graph

Five stages, fully automated — all governed by a declarative ontology that encodes your business domain.

Any data

- Snowflake · Databricks
- Oracle · SQL
- PDF · CSV · Text
- Web · APIs
- GP workflow output

→

Ontology engine

Class hierarchy & semantics

Object properties

Data properties

→

KG builder

11-phase pipeline

Structured + NL extraction

Automated · Validated

→

Knowledge graph

Neo4j database

Rheem deployment:

~20K nodes

~100K relationships

→

KG reasoner

Intelligent Q&A

Complex reasoning

Multi-agent system

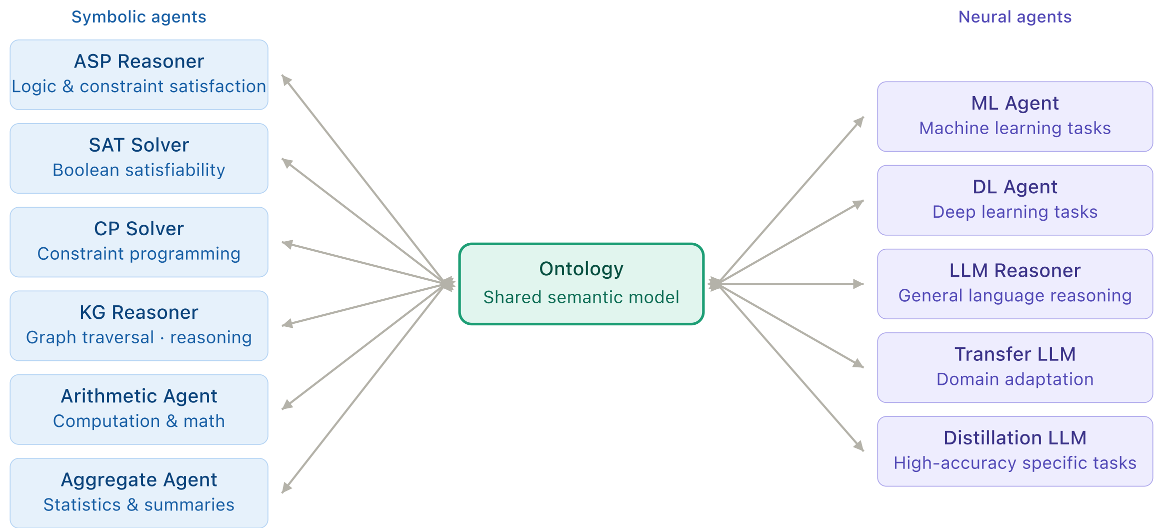
First deployment: **Rheem** — 4 business domains · 22 data partitions · up to 1.6M rows per table · 15–20K nodes · 50–100K relationships in Neo4j

GP REASONING ENGINE · SIG-PRE · 03 / 06

Multi-agent reasoning — not a pipeline, a team

All agents share a live knowledge space. The ontology decides who reasons, when, and how their conclusions become available to everyone else — instantly.

★ Two coordination modes — parallel peer sharing and ontology-driven orchestration



↔ Mode 1 — Parallel peer communication

Agents run simultaneously. Each writes its discoveries back to the Ontology — all other agents read them instantly. No central director, no handoffs, no waiting.

○ Mode 2 — Ontology-driven orchestration

The Ontology Reasoner acts as planner — deciding which agents activate, when, in what order, and how results transfer between them.

OUR COMPETITIVE ADVANTAGE · 04 / 06

The ontology — the brain behind the graph

Most systems treat all data the same. Ours understands what the data *means* for your business, and guides every agent that reasons over it.

1 It speaks your business language

Knows *SKU, Market, Manufacturer, Distribution Center, Pricing Scenario* — and their full class hierarchy. Defined once per client — zero framework changes.

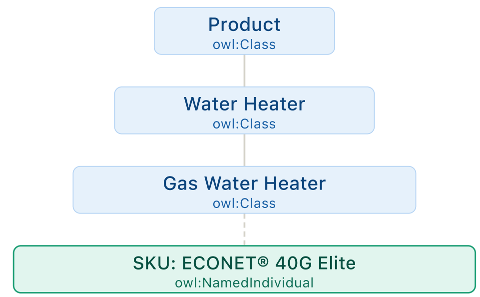
2 It ensures every insight is correctly classified

"Jacksonville" is always a *Market*, "Rheem" is always the *Manufacturer*. Every entity lands in the right place in the hierarchy — automatically.

3 It scales to new clients in days

Adding CoAction, MSL, or MSIG = one config file. The framework handles all the heavy lifting. The ontology handles all the *what matters for this client*.

Class hierarchy — Rheem product domain



owl:ObjectProperty examples

MANUFACTURED_BY → Rheem | SOLD_IN_MARKET → Atlanta | HAS_FUEL_TYPE → Gas

owl:Class | owl:NamedIndividual | - - - instanceOf

This semantic model is the backbone of every slide — it powers the KG Builder pipeline (slide 2), orchestrates all reasoning agents (slide 3), and makes every one of the 9 business capabilities on slide 5 possible.

BUSINESS VALUE · 05 / 06

What becomes possible with a knowledge graph

Connected to SIG-PRE's reasoning agents, the knowledge graph enables the full spectrum of AI reasoning over real business data.

Natural language Q&A

Ask any business question in plain English. The system traces through the graph to derive answers from known facts, or infers the most likely explanation when data is incomplete.

Deductive · Abductive · Qualitative

Competitive intelligence

Detect pricing gaps, market positioning, and competitor patterns. Generalise trends from observed market data and map analogous products to benchmark against.

Inductive · Analogical · Spatial

Supply chain & risk

Map supplier dependencies, geographic concentration, and lead times. Reason causally to propagate disruption impact before it reaches operations.

Causal · Temporal · Probabilistic · Spatial

Scenario simulation

"What if we raise prices 10% in Atlanta?" Run counterfactuals across the portfolio and find constraint-optimal outcomes before committing to any decision.

Counterfactual · Causal · Constraint-Based

Epistemic & compliance

Distinguish known facts from working assumptions. Retract conclusions when new evidence arrives and enforce obligations, permissions, and regulatory norms.

Epistemic · Default · Non-monotonic · Deontic

Multi-agent orchestration

The Meta-Reasoning layer plans which agents activate, in what order, and how results transfer — coordinating the team across complex multi-step business problems.

Meta-Reasoning · Constraint-Based · Temporal

Root cause analysis

Why are warranty claims spiking at Ontario DC? Why did Q3 margin drop? Trace backward through the graph to find the most probable cause — revise when new evidence arrives.

Abductive · Causal · Non-monotonic

Predictive analytics

Forecast demand, claim rates, or supplier lead times by generalising from historical patterns and propagating causal drivers forward along the time dimension.

Inductive · Temporal · Probabilistic · Causal

Risk scoring & prioritisation

Score and rank suppliers, products, or markets by risk. Combine probabilistic estimates, causal impact, and regulatory obligations to decide where to act first.

Probabilistic · Deontic · Causal · Inductive

LIVE DEMONSTRATION · 06 / 06

See it in action

Two windows — the ontology that drives everything, and the knowledge graph it builds.

01

Ontology in Protégé

The OWL/RDF ontology — the business vocabulary defining every entity class, object property (relationship), and data property (attribute). Our IP: the declarative schema that makes the framework client-aware.

owl:Class | owl:ObjectProperty | owl:DatatypeProperty

02

Knowledge graph in Neo4j

The live graph — 15–20K nodes, 50–100K relationships (Rheem deployment). Traverse from a SKU to its markets, competitors, pricing scenarios, warranty DCs, and suppliers — all in one connected query.

~20K nodes | ~100K links | 4 domains | Cypher queries live

What you'll see: One Rheem SKU connected to its market, competitors, pricing scenarios, warranty DCs, and supply chain — from raw data to a traversable knowledge structure, powered entirely by the ontology.